

I. Hosting Plan of Application Components

Host Site

- **Frontend (Web UI)** → Hosted on **Vercel**
- **Backend (Node.js Express API)** → Hosted on **Render**
- **Database (MongoDB)** → Hosted on **MongoDB Atlas**
- **External APIs** → Steam Web API and Gemini API (cloud hosted)

Deployment Strategy

- Push source code to GitHub repository.
- Connect **Vercel** to frontend repository for automatic deployment.
- Connect **Render** to backend repository for automatic deployment.
- Configure environment variables in Render:
 - **MONGO_URI**
 - **STEAM_API_KEY**
 - **JWT_SECRET**
- Backend connects to MongoDB Atlas using Mongoose.
- Frontend communicates with backend via REST API endpoints.
- Backend integrates Steam API and Gemini API using secure API keys.

Security Measures

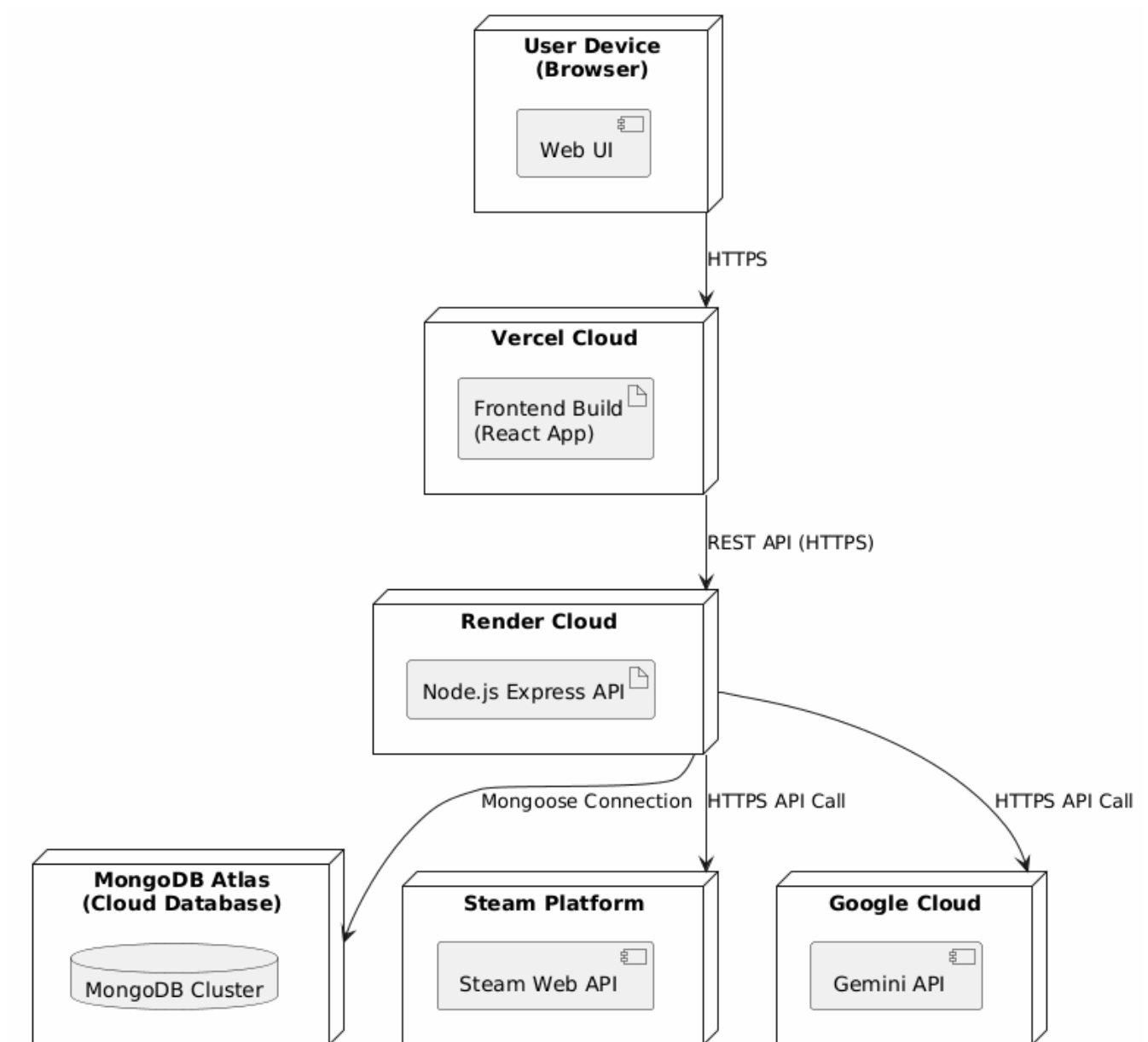
- HTTPS enabled on **Vercel** and **Render**.
- Passwords stored using **bcrypt** hashing.
- JWT authentication for protected routes.
- API keys stored securely in environment variables.
- MongoDB Atlas network access control configured.

II. End User Access and Component Interaction

How End Users Access the System

- User accesses the deployed frontend through a browser.
- User submits chatbot queries via the Web UI.
- Frontend sends HTTP request to backend API.
- Backend processes request and retrieves data from:
 - MongoDB Atlas
 - Steam Web API
 - Gemini API
- Processed response is returned to frontend and displayed to user.

Deployment / Interaction Diagram



III. Implemented Components and Their Interaction

1. Authentication Component

- Implemented user registration and login.
- Password hashing using **bcrypt**.
- JWT token generation for authentication.
- Backend validates credentials and issues secure token.

Interaction:

- User submits login request.
- Backend verifies credentials using User model.
- JWT token returned to frontend.
- Token used for accessing protected routes.

2. Steam API Integration Component

- Implemented Steam API integration using axios.
- Created endpoint to fetch owned games and playtime.
- Backend processes JSON data and formats response.

Interaction:

- User sends statistics query.
- Backend calls Steam Web API.
- Data processed and response returned to frontend.